

Speaking with avatars - influence of social interaction on movement behavior in interactive hearing experiments

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ABSTRACT

This study investigated to what extent social interaction influences the motion behavior of normal-hearing listeners in interactive hearing experiments involving audiovisual virtual reality. To answer this question, an experiment with eleven participants was performed, using two different levels of virtualization, two levels of interactivity, and two different noise levels. The task of the participants was either to communicate with two real or two virtual interlocutors (conditions 'real' and 'active') or to listen passively to a conversation between three virtual characters (condition 'passive'). During the experiment, the gaze, head and body motion of the participants was recorded. An analysis of variances showed that the gaze and saccade behavior does not differ between 'real' and 'active' conditions. Behavior was found to be different between 'active' and 'passive' conditions. For the head motion related parameters, such a significant effect was not found. A classifier was trained to distinguish between conditions based on motion and pose features. Performance was higher for the comparison between 'active' and 'passive' conditions than for the comparison between 'real' and 'active' conditions, indicating that the measured difference in motion behavior was not sufficient to distinguish between the 'real' and the 'active' condition.

Index Terms: Interactive experiments—Movement behavior—Gaze behavior—Social interaction

1 INTRODUCTION

Due to the development of new hearing aid algorithms based on the head and eye movement behavior of the user, an in-depth study of this behavior is necessary to evaluate the effectiveness of these algorithms. One way to achieve this is the use of virtual environments, which can be used to create reproducible and controllable experiments that come close to reality [9]. This type of interaction technology becomes more and more relevant and further investigated [2]. The influence of visual cues on the behavior in such a virtual situation has already been investigated [8]. To investigate also the influence of social interaction on the movement behavior in virtual audio-visual hearing experiments, the movement behavior of subjects, especially their gaze behavior, was measured in different communication situations with various levels of virtualization, interactivity and noise.

Hendrikse et al. [8] conclude that the head and eye motion behavior of the subjects is influenced by the visual condition of the experiment, specifically by the type of simulated lip and head motion. Additionally, the task performance and the subjective perception of the participants were affected by the visual cues provided.

Hadley et al. [7] investigated the speech, motion and gaze behavior of people talking in pairs in noise conditions. In noisy environments different strategies to maintain the speech intelligibility can

be found, for example by moving closer together or by adapting the speech level. Better speech intelligibility can also be achieved by lip reading, which requires to look at the speaker's mouth. The mentioned study showed that the used strategies in a real conversation in noise differ from the ones used in an empty lab. One assumption to explain these results is that regarding the speech intelligibility improvement, two factors compete with each other: the progress of the communication and the interpersonal relationship. Strategies that normally contribute to a higher speech intelligibility are avoided for social reasons. Also, the relationship of the conversational partners may have an influence on the behavior in noisy communication situations [7].

The face and especially the eyes are a very important attractor for the visual attention during a conversation. During a conversation, people look most of the time (62 %) at the person they are listening to and not at the other participants (9 %). Also, actively speaking persons look longer at the participant to whom they are talking (40 %) than at the others (12 %) [15]. This can have several reasons. The gaze direction is a very effective medium to transmit socially important information to others [13]. Through gaze focus, the relative importance of other parts of the environment, including people, can be conveyed [14]. This may be utilized to guide the attention, and in experiments with virtual audiovisual environments may affect task performance [8].

The hypothesis underlying this study is that the level of social interactivity may have a larger influence on movement behavior than the level of virtualization. By investigating the effect of social interactivity and virtualization on movement behavior in hearing experiments, this study contributes to the development of experimental methods with increased ecological validity in hearing research.

2 METHODS

2.1 Experiment

The within-subjects study consisted of an experiment for which two different levels of virtualization, two levels of interactivity and two noise levels were compared (see Table 1 for a summary). In the first condition ('real' condition) the participants communicated with two experimenters who were physically present. In the second condition ('active' condition), the participant was again communicating with the experimenters, but now in a virtual audiovisual environment. In this condition, the experimenters were located in a different room, and only the voice was transmitted. The visual appearance was rendered using virtual characters with simulated lip- and head-movements. In the third condition ('passive' condition) the participant listened to a pre-recorded conversation, again reproduced with animated virtual characters in an audio-visual environment. Each condition was measured twice, with and without diffuse cafeteria background noise. The noise signal was reproduced at 65 dB SPL (unweighted) using eight distributed loudspeakers at ear level in laboratory 1 ('real' condition) and using 45 loudspeakers spherically distributed around the listener in laboratory 2 ('active' and 'passive' conditions). There were no restrictions on the topics of the conversation, a set of 30 different pictures showing everyday situations could

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Table 1: Summary of test conditions. All conditions were measured in silence and in noise.

condition	'static'	'real'	'active'	'passive'
virtualization	–	real	virtual	virtual
interactive	no	yes	yes	no
movement allowed	no	yes	yes	yes
laboratory	1 and 2	1	2	2

be used as an inspiration to keep the conversation going. Eleven normal-hearing subjects (6 female, 5 male, aged between 22 years to 57 years) participated in this task. The study was approved by the Research Ethics Committee of the University of Oldenburg.

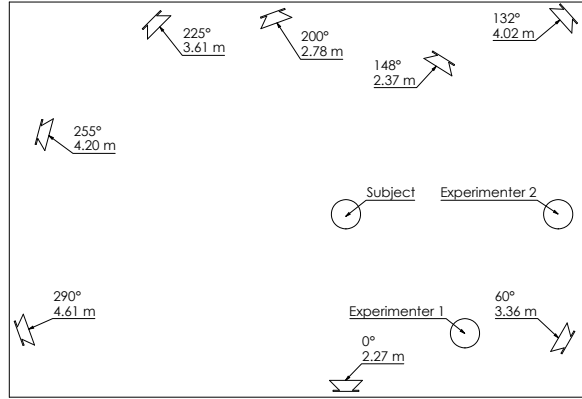


Figure 1: Schematic top view of the loudspeaker and speaker positions in the laboratory 1.

The 'real' condition was measured in the first lab (see Fig. 1 for an overview of the arrangement of participants, experimenters and loudspeakers). The participant was sitting in front of the two experimenters. To record the participant's face, a Canon EOS 700D camera with a EFS 18-55 mm lens was used. For the measurement of the body movement a Microsoft kinect depth camera was installed. The speech of the participants was recorded with a microphone (type Neumann KM 183) next to the participant. Both experimenters wore a condenser lavalier microphone (type AKG C 417 PP) directly attached to their body. Because of the restrictions due to the Covid-19 pandemic, the test participants wore a transparent face shield and both experimenters masks. The influence of such a face shield on the analysis with the tool OpenFace was investigated before the experiment. No major effect on the reliability of gaze estimation was found. The 'active' and the 'passive' condition were measured in a second lab (see Fig. 2). Here, the participant is sitting alone in a 'tent' with a cylindrical, acoustically transparent screen of 3.52 m diameter, on which the visual scene can be projected on (see Fig. 3). The scene represents a virtual version of the first lab with the female avatars sitting in the same position as the experimenters and the participant also sitting in the same position as in the 'real' condition (see Fig. 4). The videos of the participant's face and the movement are made with the same cameras as in the first lab in the same positions relative to the participant. Here, the speech of the participant was recorded with a condenser microphone type AKG C 417 PP hanging from the roof of the laboratory.

A 'static' reference condition was measured in both laboratories. In this condition the participant was not moving and looking to the front for one minute.

In two conditions, the participant sits in this lab while the two experimenters are communicating with the participant using avatars,

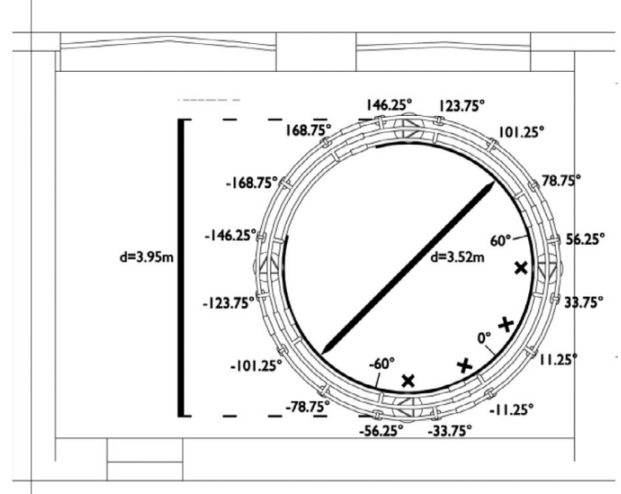


Figure 2: Schematic top view of the loudspeaker positions in the laboratory 2 [8].



Figure 3: Photo of the actual setup showing the loudspeakers (blue circles)

which are projected onto the screen. To transmit the speech of the experimenters, they wore a headset (type AKG HSD 271 Studio Set), which is connected to the lab via network using an 'ovbox' [3] (Raspberry Pi 4B Computer with a Focusrite Scarlett solo audio interface). This way, the participant hears the two experimenters via loudspeaker, but it seems like they are sitting in the same position as the avatars.

The virtual characters [12] were always moving their head towards the active speaker, regardless if it is a virtual or a real person, and their size is adjusted to the size of a real human sitting in the same position as the avatar in the real room¹. For their mouth movement, a speech-driven lip-syncing that uses a weighted average between three blend shapes was used. Using inverse kinematics, the avatars could move their head and eyes. They blinked every 0.1 to 4 s with a duration of 0.15 to 0.5 s. Gestures were not included in the avatar movement.

For the acoustical scenes and the feature recording of the kinect camera, the Toolbox for creation and rendering of dynamic virtual

¹The avatars were already implemented and imported into the scene. Their automated behavior is further described in a work by Hendrikse et al. [8]



Figure 4: Virtual scene that is projected onto the screen in the tent

acoustic environments (TASCAR, [6]) was used. This toolbox allows the creation of a reproducible virtual acoustic environment, where the configurations of the incoming audio signal and the acoustical features of the scene are known for each point in time.

The microphone signals of the participants and of the experimenters were recorded for further analysis.

2.2 Analysis

The recorded audio was annotated manually and used to separate the movement measures into all possible speech activity states. For the analysis, only the single-speaker-states were relevant. Two Machine Learning tools were used to analyse the video data, OpenFace and a naive Bayesian classifier. The OpenFace tool was used to determine the gaze angle and head rotation of the participants. By receiving a video or a picture as an input, the tool automatically detects facial landmarks and estimates the gaze direction [1]. Then, the mean gaze angle and the mean head rotation angle of each participant were calculated for each single-speaker state. Additionally, by taking the temporal derivative to get the gaze speed in $^\circ/s$ and by determining a threshold value of $50^\circ/s$, the total number of saccades could be obtained. In every frame, in which the temporal derivative is bigger than the threshold, a saccade was measured. If this was the case for two or more consecutive frames, only one saccade was counted. The counted saccades were then divided by the total duration of the speaker state in the investigated condition to remove the temporal dependence of the number of saccades from the duration of the sequence and the final measure of interest was the saccade frequency. Also, by taking the temporal derivative of the head rotation angle, the head rotation speed was determined and the RMS of the speed was measured.

To determine if the four measures differ significantly regarding several aspects, various analyses of variance ANOVAs with repeated measurements were performed. For each measure it was investigated, whether the virtualization condition, the noise condition or the interactivity (passive vs. active communication) have a significant effect on the movement data. If this was the case for one or more aspect, post-hoc tests were executed to figure out which subconditions were significant. Additionally, a gesture classification algorithm was used to identify the test condition based on movement and posture features extracted from the marker free depth camera [5]. Half of the participants were used to train this naive Bayesian classifier, and the other half was used to calculate the classification performance. A bootstrapping method was applied to achieve multiple samples of the classification performance. The classifier was adapted to this experiment by discarding the features that lead to a high classification performance in a static condition, where in both labs the participant sits on a chair without moving and looks directly into the camera. Then, the 'real' condition was compared to the 'active', virtual condition. Also, the two virtual conditions, where the participant is

one time active and one time a passive listener to a conversation between three avatars were contrasted. For the purpose of testing the influence of noise on the classification and therefore on the body movement behavior, the data of the situation with and without noise was used as an input of the classifier for each condition, resulting in three more combinations.

For each of the data combinations named above, the mean performance of the classifier and the standard deviation after ten repetitions of each classification were extracted for a comparison.

3 RESULTS

3.1 Gaze-related parameter

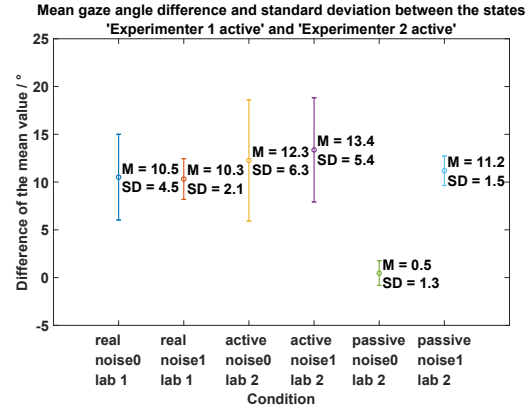


Figure 5: Mean gaze angle difference and standard deviation for the speaker states *Experimenter 1 active* and *Experimenter 2 active* for all participants and all conditions, i.e., the amount of head movement caused by turn-taking.

The mean difference and standard deviation between the mean gaze angles of all participants for the two possible speaker states *Experimenter 1 active* and *Experimenter 2 active* for all conditions are depicted in Fig. 5. This difference corresponds to the effect of turn-taking on the gaze angle. Except for the 'passive' condition without background noise, all mean values lie within the range of 10.3° - 13.4° . The 'real' condition without noise leads to a mean gaze angle difference of 10.5° with a standard deviation of 4.5° . The mean difference for the 'real' condition with noise (10.3°) is similar to the condition without noise, but the standard deviation (2.1°) is smaller. The 'active' condition without noise leads to a mean difference of 12.3° . Here the standard deviation (6.3°) is highest, in comparison to the other conditions for this speaker state combination. Also in the 'active' condition with noise, the standard deviation (5.4°) is high, but the mean difference of 13.4° is similar to the other values already named. The 'passive' condition without noise with a mean value of 0.5° and a standard deviation of 1.3° differs from the other ones. The 'passive' condition with noise with a mean difference of 11.2° and a standard deviation of 1.5° is in the same order as the 'real' and the 'active' conditions.

At the significance level of 0.05, the ANOVA shows that there was no significant effect of the three conditions 'real', 'active' and 'passive' on the mean gaze angle, $F(2, 16) = 3.04$, $p = 0.076$, $\eta_p^2 = 0.28$. Still, there was an effect of the noise condition on the behavior, $F(1, 8) = 17.04$, $p = 0.003$, $\eta_p^2 = 0.681$. This difference was then further investigated regarding the three conditions and led to the result that only in the 'passive' condition, the noise condition results in a significant difference of the mean gaze angle, $p < 0.001$. Additionally, the active speaker had a significant impact, $F(2, 16) = 103.62$, $p = 0.000$, $\eta_p^2 = 0.93$ in all possible speaker combinations. Further

tests of this regarding the noise condition showed that this behavioral difference is significant for all speaker comparisons and all noise conditions except for the three combinations in the 'passive' condition without noise. There, the mean gaze angle did not differ significantly, depending on the active speaker. The effect of the condition regarding the speaker in the two noise conditions was not significant, if the 'real' and the 'active' condition were compared to each other. The 'active' versus the 'passive' condition lead to a significant difference for the second experimenter in both noise conditions and for the first experimenter only without noise.

Another ANOVA showed that for the saccade frequency, the three conditions 'real', 'active' and 'passive' did also not have a significant effect, $F(2, 16) = 0.106$, $p = 0.900$, $\eta_p^2 = 0.013$. Unlike as for the mean gaze angle, the noise condition also did not have a significant impact, $F(1, 8) = 0.001$, $p = 0.979$, $\eta_p^2 = 0.0$. Only the different speakers had a significant effect, $F(2, 16) = 18.107$, $p < 0.001$, $\eta_p^2 = 0.694$. This is again the case for all speakers in all conditions. A further analysis regarding the noise condition revealed that the difference for the active speaker was significant for all conditions with and without noise, except for the 'passive' condition without noise. Here the saccade frequencies did not differ significantly depending on the active speaker. The comparison of the conditions regarding the active speaker showed that only if the 'active' and the 'passive' condition are compared, the values differed significantly. This was the case for the first and the second experimenter in the condition without noise, but not for the first experimenter with noise.

3.2 Head-related parameter

Regarding the three conditions 'real', 'active' and 'passive', the performed ANOVA showed that there was no significant effect of the head rotation angle, $F(2, 16) = 0.504$, $p = 0.613$, $\eta_p^2 = 0.059$. Additionally, neither the noise condition, $F(1, 8) = 3.42$, $p = 0.101$, $\eta_p^2 = 0.300$, nor the active speaker, $F(2, 16) = 0.032$, $p = 0.96$, $\eta_p^2 = 0.004$, had a significant effect. Therefore no further analyses were performed.

A last ANOVA showed that the condition did not have an effect on the RMS of the head rotation speed, $F(1.23, 9.88) = 0.963$, $p = 0.371$, $\eta_p^2 = 0.107$. Furthermore, the noise condition, $F(1, 8) = 2.872$, $p = 0.129$, $\eta_p^2 = 0.264$, and the active speaker, $F(1.14, 9.16) = 0.614$, $p = 0.475$, $\eta_p^2 = 0.071$, did not effect the measured values significantly.

3.3 Classification performance

The 'static' condition which was used to adapt the classifier to the experiment lead to a mean performance of 66.8 %, see Fig. 6, leftmost bar. In this condition where the participants were sitting without movement on a chair looking directly into a camera, for the two labs was compared to each other.

The classification between the 'real' and the 'active' condition when the participant was an active part of the conversation lead to a mean performance of 65.1 %. As these two conditions were measured in the first and the second lab, respectively, a performance similar to the 'static' condition in which only the lab differed means that no effect of 'real' versus 'active' condition on movement behavior was found.

The classification of the 'active' and the 'passive' condition, which were both measured in the second lab, resulted in a mean performance of 61.4 %. Here, a performance larger than chance level of 50 % indicates an effect of condition on the movement behavior.

To test the influence of noise, the classifier received the data with and without background noise as an input for all conditions. For the 'active' condition, the mean performance was then 56.3 %, for the 'passive' condition 54.4 % and for the 'real' condition 59.2 %. Again, values above chance level indicate an influence of noise

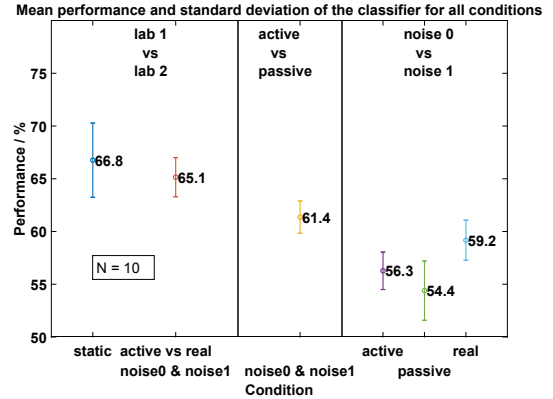


Figure 6: Mean performance and standard deviation of the classifier for six combinations of conditions. In the first two conditions the classifier was trained to distinguish between the two laboratories. In the first condition the participants were asked not to move; an effect of the level of virtualization on motion behavior would have resulted in a higher performance in the second condition than in the reference condition. For the remaining four conditions the chance level serves as a reference. The level of interactivity has an effect on movement behavior (third condition), and also the noise level (fourth to sixth condition).

condition on movement behavior. This influence was largest in the 'real' condition and smallest in the 'passive' condition.

4 DISCUSSION

Despite some limitations, the data clearly suggest that for a larger ecological validity of motion behavior data in hearing experiments with virtual audiovisual environments, an interactive paradigm may be required. This is in line with the requirements towards a larger ecological validity in [11], and it supports the findings of a recent review on self-motion [4], which, in spite of the relevance of motion behavior for hearing-device function, found only very little research on interactivity and self-motion in the context of hearing and hearing aid research. As a consequence, it might be desired to find experimental paradigms and tasks which increase the level of interactivity without weakening the sensitivity and reproducibility of established methods. Towards such paradigms the model of a communication loop as defined in [10] may provide a guideline, as it formalizes the building blocks and their interaction.

In addition to the importance of interactivity in hearing experiments the data show that using virtual animated characters instead of real physically present interlocutors is possible, at least as long as a certain degree of interactivity is provided. Nevertheless there are some limitations of the experiment. The face visor worn by the participants due to the Covid-19 restrictions may have influenced their behavior. The speech sounds different, and the fact that one always feels something attached to the head might influence the speech- and motion behavior. Therefore, the results might be different than they would have been without the visor. Especially the fact that the visor is only worn in the 'real' condition, as in the other conditions the participant is alone in the lab, eventually leads to different results for the comparison between the 'real' and the 'active' condition. This also leads to the assumption that a measurement with the participant and the two experimenters all sitting in the tent, and talking directly to each other would be a better 'real' condition, as the interlocutors would be real, but the background, and therefore the scene, would be the same for all three conditions. However this measurement was also not possible due to the restrictions, as it is not possible to

maintain a distance of 1.5 m in between all interlocutors in the tent and only two persons were allowed at the same time in the lab. These restrictions also led to a limited number of participants as no external participants were allowed. Additionally, the influence of the masks worn by the experimenters and the difference to the avatars who did not wear masks on the behavior of the subject was not investigated. Secondly, the speech signals for the 'passive' condition, i.e. the conversation between the three avatars, was recorded without any background noises – this was added later on. Therefore, the SNR varies strongly in the noise condition. This can have a big impact on the speech intelligibility and consequently also on the movement behavior. The pre-recording of the conversation also took place in three different rooms with the interlocutors using headsets to enable recording without masks. This also means that their conversation might not be completely natural and this may have influenced the reactions of the participant. A repetition of the experiment when the restrictions do not have to be maintained would be necessary in order to validate the results mentioned above.

Furthermore, the movement behavior of the avatars is not fully natural, especially in the conditions with background noises. The levels sometimes did not fall below the threshold used to trigger gaze changes, thus the avatars did not look to the participant, although he or she was active. Therefore the participants sometimes potentially expect another behavior of the avatar and also react differently than they would if the avatars were more natural. Thus, an improvement of the avatar behavior or an experiment with an adapted threshold or the use of an contact microphone for the participants, which was again not possible due to the Covid19 restrictions, can help to improve the results.

Regarding the task of the participants and the conversation between the three avatars, there were no constraints on the topics and time in which the participant is active. But this may lead to very variable topics and speech duration. Some participants talk longer and are more passionate about some of the topics. This potentially influences the motion behavior. Therefore the development of a more limited task, which still leads to a normal conversation, might be helpful. This eventually helps to induce similar speaker durations and avoid the influence of the emotional differences between speakers and topics, but on the other hand may introduce a bias due to a lower level of realism. Additionally, the experimenters were asked to behave as they would in everyday life in such a situation. Therefore, no restrictions of their movement and gesture behavior were given, as the goal was a complete natural conversation between them and the participant. Nevertheless, the influence of the difference in motion behavior of the real experimenters and the virtual avatars on the behavior of the subject was not investigated.

5 CONCLUSIONS

Data suggest that the level of interaction and the noise level may influence the motion behavior of normal-hearing listeners in interactive hearing experiments involving audiovisual virtual reality. The level of virtualization showed a smaller effect on motion behavior; this effect was not significant for the gaze behavior.

These findings suggest that care should be taken when designing hearing experiments with virtual reality, which utilize or assess gaze or motion behavior: The gaze and motion behavior in a passive listening environment significantly differed from a real conversation with physically present interlocutors, which from the tested conditions can be seen as the ecologically most valid condition. The effect of using a condition with virtual characters only marginally affected motion behavior as long as the conversation was truly interactive. This indicates that a high level of ecological validity can be achieved for movement behavior even with virtual reality.

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